REALM HACK

ThinkFirst

TEAM

Odysseus

TRACK

Agentic AI

MEMBERS

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Problem Statement

- Overreliance on AI: Students use chatbots to get answers, skipping the effort that builds understanding.
- Hollow learning: Critical thinking and subject knowledge decline while grades stay high.
- One-size-fits-all answers: Chatbots ignore a student's level and hand over solutions instead of teaching.
- Educator blind spot: Teachers can't see who struggles with what, so help arrives late.
- The gap: No tool guides students toward understanding while showing teachers exactly where learning breaks down

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Solution & Objectives

Proposed Solution

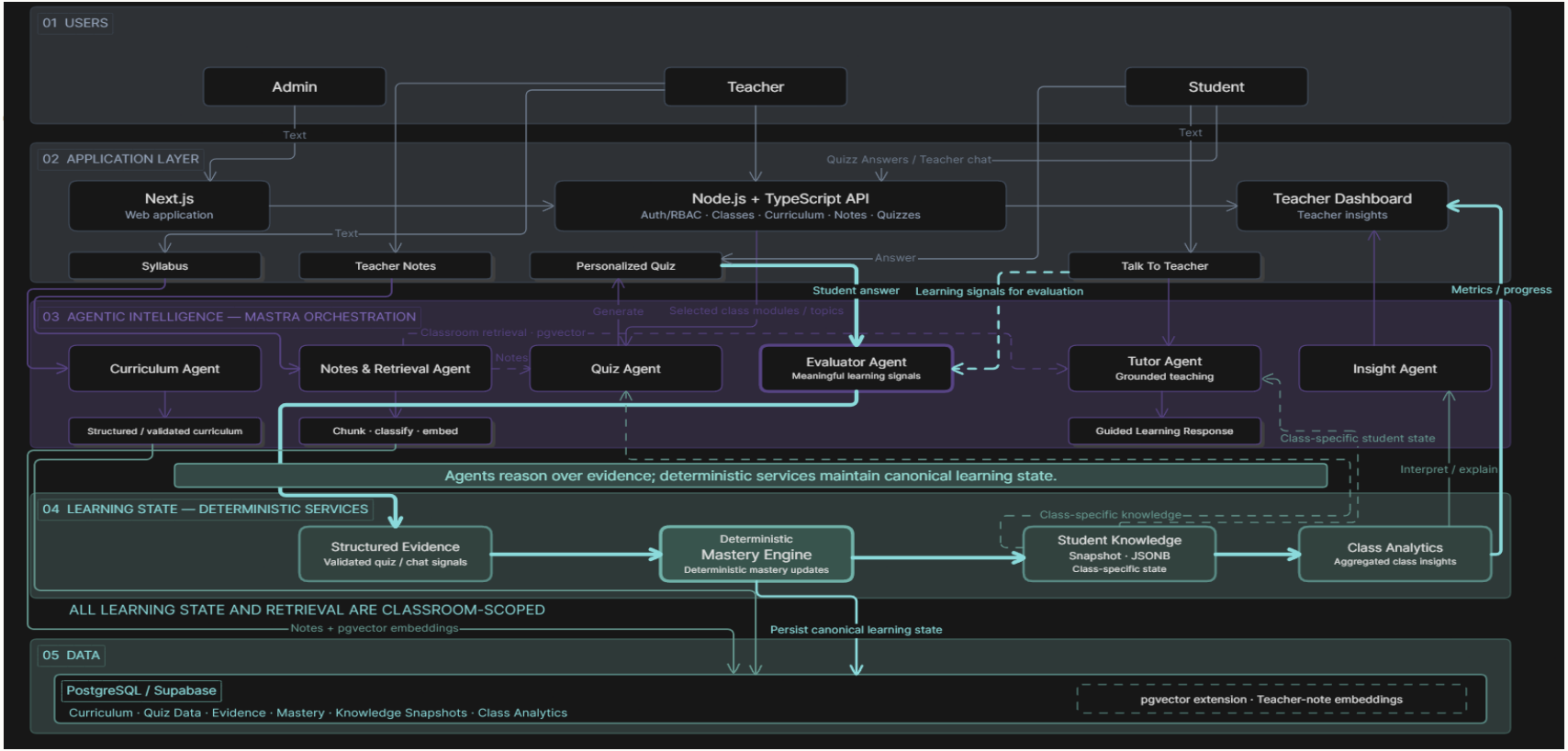
- **AI Teacher Agent :** Acts like a personal tutor, helping students learn step by step instead of simply providing answers.
- **Student Understanding Tracker :** Monitors what each student knows, struggles with, and improves over time.
- **Teacher Analytics Dashboard**: Real-time visibility into specific class learning gaps

Project Objectives

- **Promote Active Leaning**: Eliminate answer-copying via step-by-step guided reasoning.
- **Eliminate Educator Blind Spots**: Provide immediate signals when student comprehension breaks down.
- **Ensure Syllabus Grounding**: Constrain AI responses strictly to teacher materials via pgvector RAG

System Architecture

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Tech Stack

Layer	Technologies	Purpose
Frontend	Next.js · Tailwind CSS v4 · shadcn/ui	Student, Teacher & Admin Web App
Backend	Node.js · TypeScript · Express.js	REST APIs & Core Services
Authentication	Supabase Auth · RBAC	Secure Role-Based Access
Agentic AI	Mastra · Mastra Workflows · Gemini 3.8 Flash	Curriculum, RAG, Quiz, Evaluation, Tutor & Insights
Data	PostgreSQL · Supabase · JSONB	Learning State, Curriculum & Evidence
Retrieval	pgvector · Gemini Embedding 2	Semantic Search & Class-Scoped RAG
Intelligence	Structured JSON · Evidence Extraction	Personalized Assessment & Grounded Teaching

We confirm this project will be built during the hackathon, not before.

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